

Wind farm



Bus 1 · offshore AC



Converter 1 · grid-forming
 P/ω droop: $id_{ref} \leftarrow P_0 + \Delta P + K_{wi} \cdot \Delta \omega$
QV droop: $iq_{ref} \leftarrow \beta V \cdot \Delta Q$



id_{c1}, v_{dc1}

DC cable · R_{dc}, H_{dc1}, H_{dc2}



id_{c2}, v_{dc2}

Converter 2 · grid-following
 V_{dc} control: $id_{ref} \leftarrow K_{pd} \cdot \Delta V_{dc} + K_{id} \cdot \int \Delta V_{dc}$
Q/Vac droop: $iq_{ref} \leftarrow \alpha_2 \cdot \Delta Q + \beta_2 \cdot \Delta V_{ac}$



Bus 2 · onshore AC grid